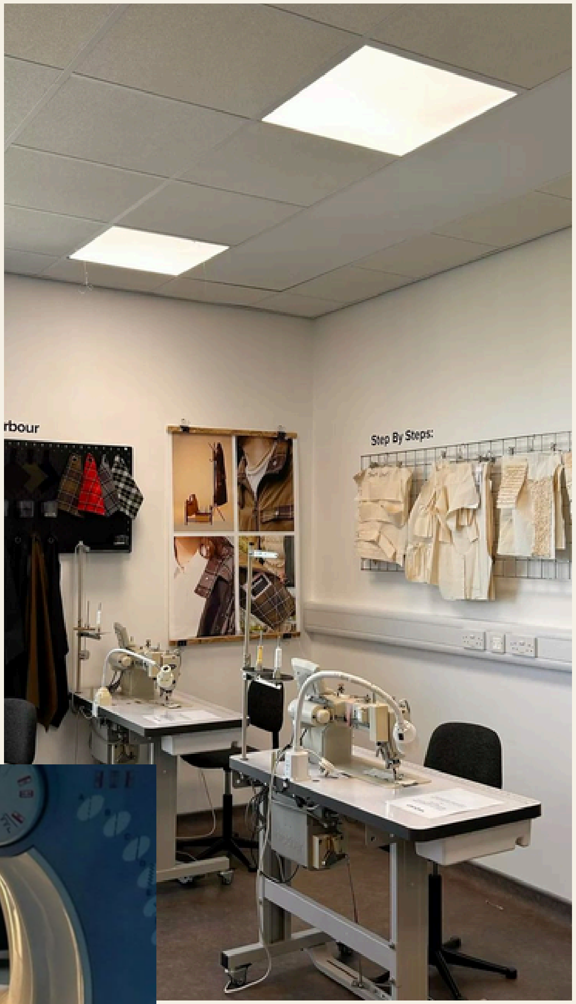
ID 1803- Section I
Spring 2025

TAILOR-D

TAILOR-D is the step stool entirely tailored to you, so that whether a designer or customer, a railing, step, or seat is always readily accessible.

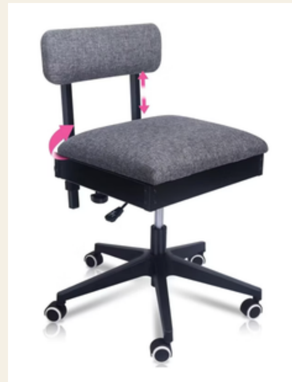
Sophie Roach & Melissa Yao



- Industrial
- Geometric
- Vintage

Edge Treatment:

Half Bullnose

Product & Brand	WAWAK	Pink Power Sewing Chair	BEKVÄM Step Stool	Hltdzer Convertible Step Stool
Image				
Cost	$158.50	$389.99	$59.99	$168.12
Material	Wood	Stainless steel and upholstered fabric	Beech	Pine
Dimensions	30" x 6'	20' x 20'	17' x 25.75'	14.75' x 35'
Users	Tailors- for clients to stand on so that measurements are easily taken	Made to use for sewing assistance; other crafters	General use; seems to mainly be for the average homeowner	General use; for the average person prioritizing efficiency and storage
Features	Open, accessible, carpeted, durable wood	Adjustable height, storage, wheels, cushion	Foldable, durable material in wood	Convertible, foldable, durable material in wood

Must Have

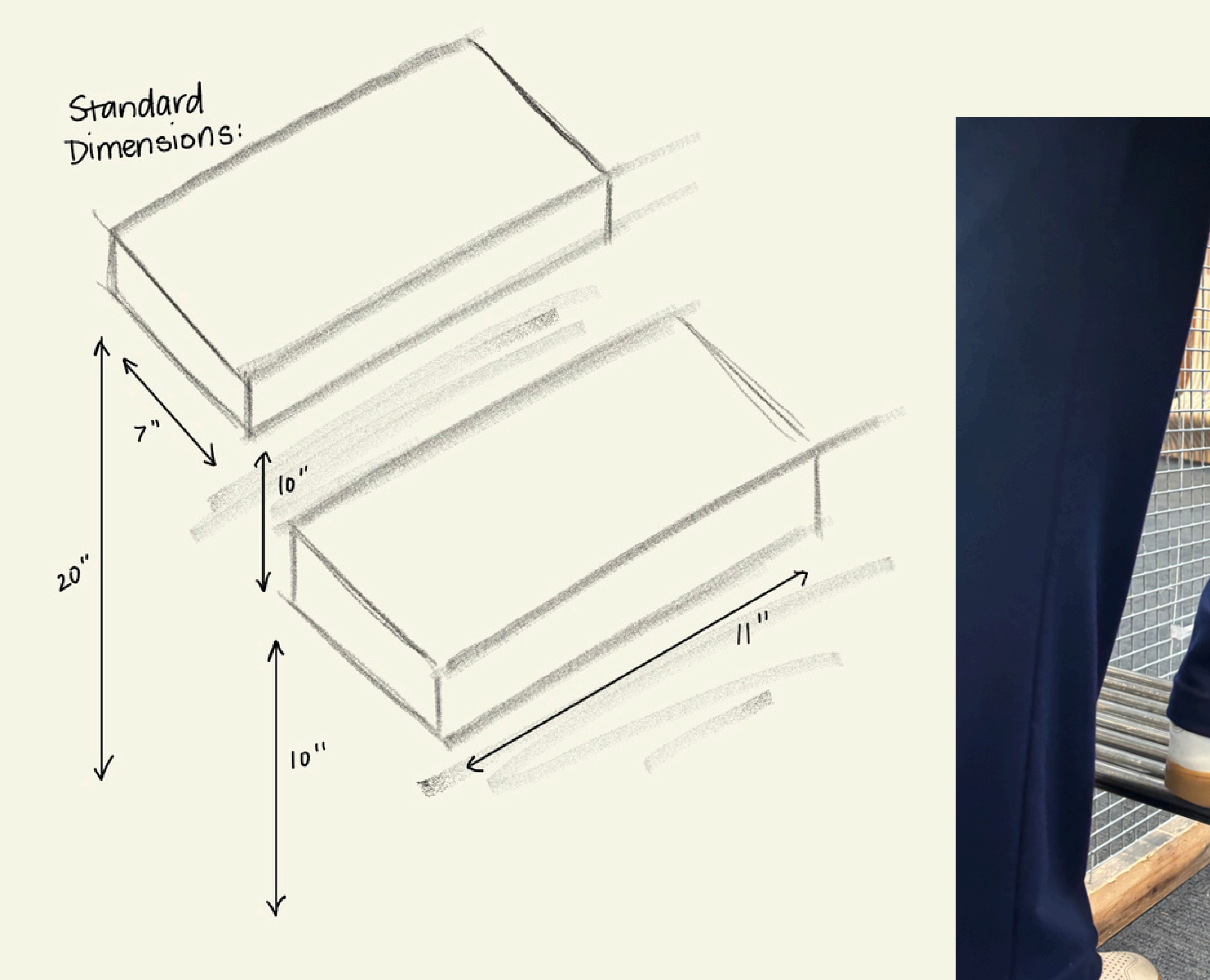
- Open and accessible from many angles
- Multi-functional
- Durable material- like wood
- Solid form to support material and function

Could Have

- Increased height to compensate for height differences between the client & tailor
- Backing support piece for clients to pose and rest while being measured
- Foldable

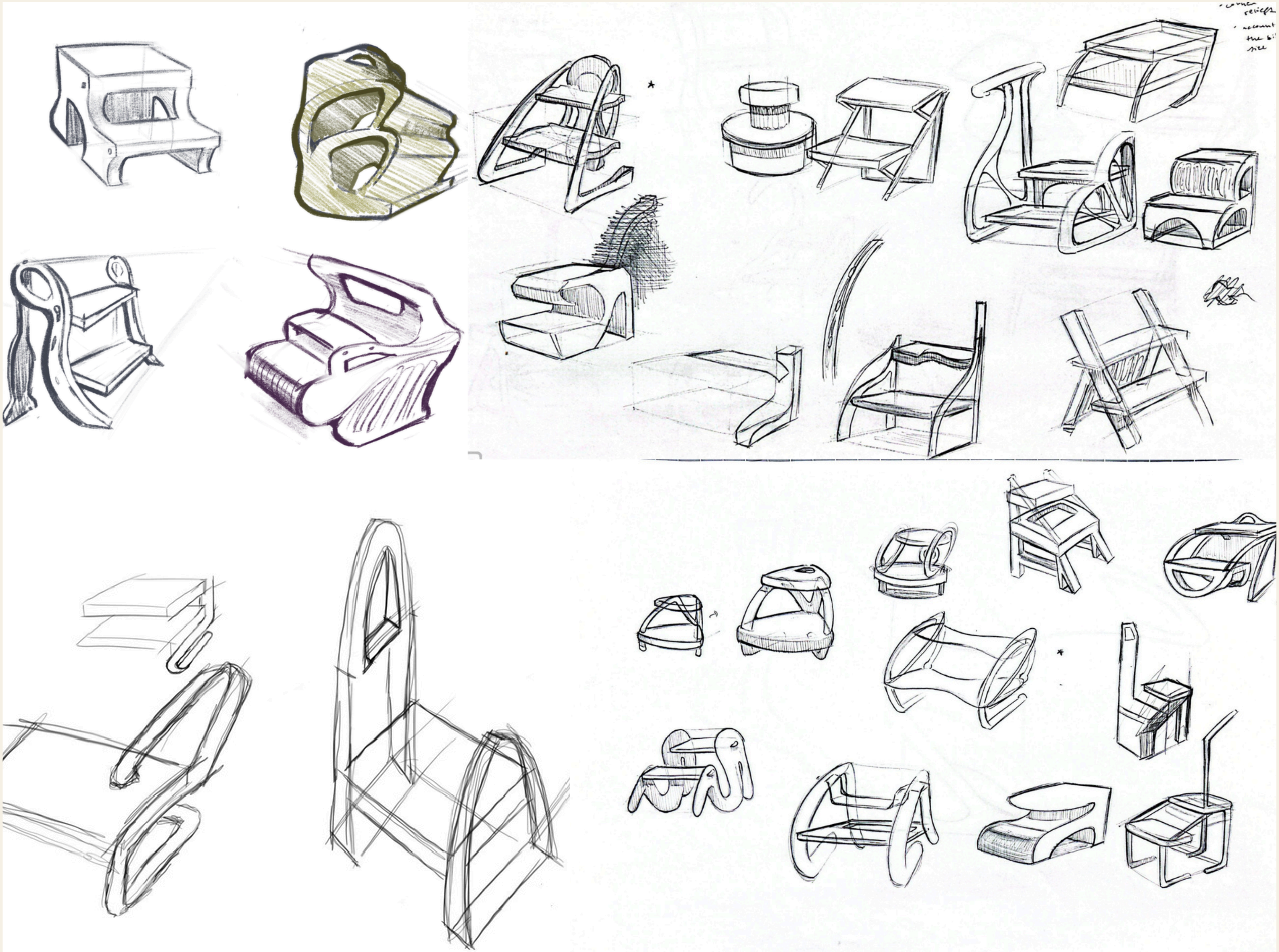
Not Feasible

- Adjustable height
- Industrial material
- Cushioned with supplementary fabric or material



CAGE Testing & Results:

We found that between the both of us, anywhere between an 8" to 10" inch step height was perfect. For our stool we ended up going with 10" to help with proportionality against the backing. We also chose a 7" step width to allow enough room for the foot & still help us stay within the material constraints.


reliefs
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size

¼ Scale Iterations

Wanted to experiment with height, curvature vs. geometric form, and the functionality of different support pieces

We chose to move forward with this back iteration as we liked the design of the support pieces & the crossed, high backing reminded us of men's tailoring.

½ Scale Iteration

We wanted to build off of our initial ¼ scale model in terms of keeping the aesthetic of all elements, but decided to enlarge the triangular sections of the backing piece for more openness, attach the bottom legs to the backing, and adjust all front support between the steps to have identical angles.

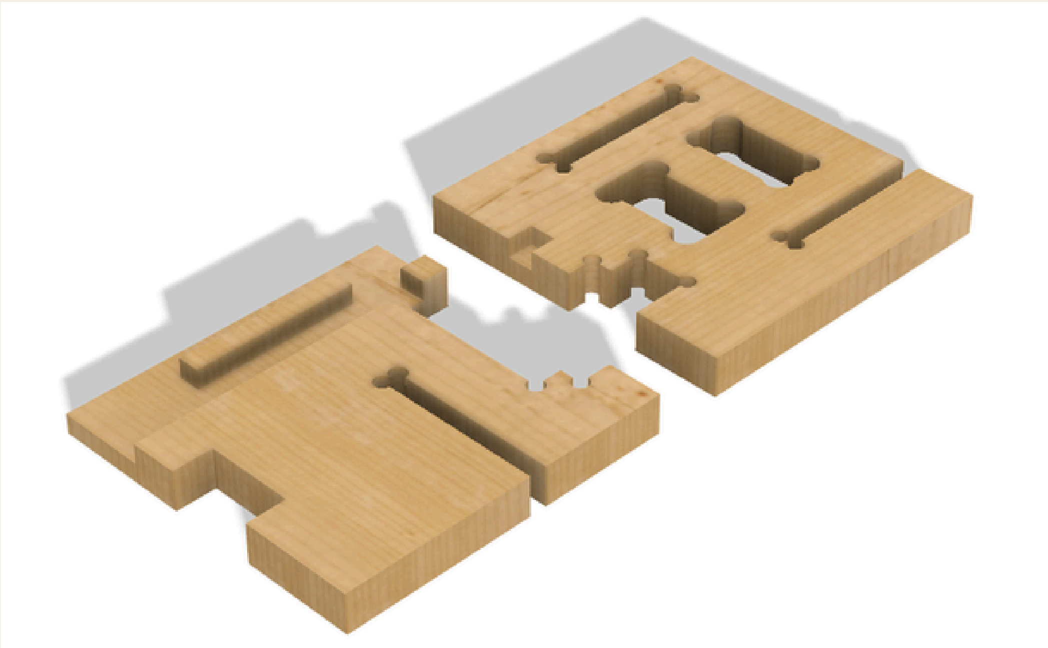
Full Scale Model

We loved everything about this model but realized we needed an additional support piece between the two sides.

Added step support from the back using an angular joint

Connected the bottom of the backing piece for stability

Fusion Test Joinery File:



T-Bone Joint Test- Full Scale

Dog-Bone Joint Test- Full Scale

CNC Cutting & File

Once we realized we needed additional stability between the two sides we decided to add a domino joint below the bottom step. This is why an extra piece is featured on our CNC file and final product.